Lora 实战

Step1 导入相关包

```
from datasets import Dataset
from transformers import AutoTokenizer, AutoModelForCausalLM,
DataCollatorForSeq2Seq, TrainingArguments, Trainer
```

Step2 加载数据集

```
import shutil
import os

# 源路径（input目录下的文件/文件夹）
source_path = "/kaggle/input/alpaca-data-zh/alpaca_data_zh/"

# 目标路径（working目录下，可自定义名称）
target_path = "/kaggle/working/alpaca_data_zh/"

# 若目标路径已存在，先删除（可选）
if os.path.exists(target_path):
    shutil.rmtree(target_path)  # 删除文件夹
    # 若复制单个文件，用 os.remove(target_path)

# 复制文件夹（包括所有子文件）
shutil.copytree(source_path, target_path)

# 若复制单个文件，用：
# shutil.copyfile(source_file, target_file)

# 验证是否复制成功
print("复制后的文件列表：", os.listdir(target_path))


ds = Dataset.load_from_disk("/kaggle/working/alpaca_data_zh/")
ds
```

```
复制后的文件列表： ['dataset_info.json', 'data-00000-of-00001.arrow',
'state.json']

Dataset({
    features: ['output', 'input', 'instruction'],
    num_rows: 26858
})
```

```
ds[:3]
```

```
{'output': ['以下是保持健康的三个提示：\n\n1. 保持身体活动。每天做适当的身体运动，如散步、跑步或游泳，能促进心血管健康，增强肌肉力量，并有助于减少体重。\n\n2. 均衡饮食。每天食用新鲜的蔬菜、水果、全谷物和脂肪含量低的蛋白质食物，避免高糖、高脂肪和加工食品，以保持健康的饮食习惯。\n\n3. 睡眠充足。睡眠对人体健康至关重要，成年人每天应保证 7-8 小时的睡眠。良好的睡眠有助于减轻压力，促进身体恢复，并提高注意力和记忆力。',
  '4/16等于1/4是因为我们可以约分分子分母都除以他们的最大公约数4，得到（4÷4）/ (16÷4）=1/4。分数的约分是用分子和分母除以相同的非零整数，来表示分数的一个相同的值，这因为分数实际上表示了分子除以分母，所以即使两个数同时除以同一个非零整数，分数的值也不会改变。所以4/16 和1/4是两种不同的书写形式，但它们的值相等。',
  '朱利叶斯·凯撒，又称尤利乌斯·恺撒（Julius Caesar）是古罗马的政治家、军事家和作家。他于公元前44年3月15日被刺杀。 \n\n根据历史记载，当时罗马元老院里一些参议员联合起来策划了对恺撒的刺杀行动，因为他们担心恺撒的统治将给罗马共和制带来威胁。在公元前44年3月15日（又称“3月的艾达之日”），恺撒去参加元老院会议时，被一群参议员包围并被攻击致死。据记载，他身中23刀，其中一刀最终致命。'],
 'input': ['', '输入：4/16', ''],
 'instruction': ['保持健康的三个提示。', '解释为什么以下分数等同于1/4', '朱利叶斯·凯撒是如何死亡的？']}
```

```
from huggingface_hub import snapshot_download

# 仓库地址（如 "Langboat/bloom-1b4-zh"）
repo_id = "Qwen/Qwen3-0.6B"
# 下载路径（自定义，如 "./bloom-1b4-zh"）
local_dir = "./Qwen/Qwen3-0.6B"

# 下载整个仓库
snapshot_download(
    repo_id=repo_id,
    local_dir=local_dir,
    # 可选：指定分支（默认 main）
    revision="main",
    # 可选：强制覆盖已有文件
    force_download=True
)
```

```
Fetching 10 files:   0%|          | 0/10 [00:00<?, ?it/s]
config.json:   0%|          | 0.00/726 [00:00<?, ?B/s]
README.md: 0.00B [00:00, ?B/s]
merges.txt: 0.00B [00:00, ?B/s]
LICENSE: 0.00B [00:00, ?B/s]
generation_config.json:   0%|          | 0.00/239 [00:00<?, ?B/s]
.gitattributes: 0.00B [00:00, ?B/s]
tokenizer_config.json: 0.00B [00:00, ?B/s]
model.safetensors:   0%|          | 0.00/1.50G [00:00<?, ?B/s]
```

```
tokenizer.json:   0%|          | 0.00/11.4M [00:00<?, ?B/s]
vocab.json: 0.00B [00:00, ?B/s]
'/kaggle/working/Qwen/Qwen3-0.6B'
```

```
!pwd
```

```
/kaggle/working
```

Step3 数据集预处理

```
tokenizer = AutoTokenizer.from_pretrained("./Qwen/Qwen3-0.6B")
tokenizer
```

```
Qwen2TokenizerFast(name_or_path='./Qwen/Qwen3-0.6B', vocab_size=151643,
model_max_length=131072, is_fast=True, padding_side='right',
truncation_side='right', special_tokens={'eos_token': '<|im_end|>',
'pad_token': '<|endoftext|>', 'additional_special_tokens': ['<|im_start|>',
'<|im_end|>', '<|object_ref_start|>', '<|object_ref_end|>',
'<|box_start|>', '<|box_end|>', '<|quad_start|>', '<|quad_end|>',
'<|vision_start|>', '<|vision_end|>', '<|vision_pad|>', '<|image_pad|>',
'<|video_pad|>']}, clean_up_tokenization_spaces=False,
added_tokens_decoder={
    151643: AddedToken("<|endoftext|>", rstrip=False, lstrip=False,
single_word=False, normalized=False, special=True),
    151644: AddedToken("<|im_start|>", rstrip=False, lstrip=False,
single_word=False, normalized=False, special=True),
    151645: AddedToken("<|im_end|>", rstrip=False, lstrip=False,
single_word=False, normalized=False, special=True),
    151646: AddedToken("<|object_ref_start|>", rstrip=False, lstrip=False,
single_word=False, normalized=False, special=True),
    151647: AddedToken("<|object_ref_end|>", rstrip=False, lstrip=False,
single_word=False, normalized=False, special=True),
    151648: AddedToken("<|box_start|>", rstrip=False, lstrip=False,
single_word=False, normalized=False, special=True),
    151649: AddedToken("<|box_end|>", rstrip=False, lstrip=False,
single_word=False, normalized=False, special=True),
    151650: AddedToken("<|quad_start|>", rstrip=False, lstrip=False,
single_word=False, normalized=False, special=True),
    151651: AddedToken("<|quad_end|>", rstrip=False, lstrip=False,
single_word=False, normalized=False, special=True),
    151652: AddedToken("<|vision_start|>", rstrip=False, lstrip=False,
single_word=False, normalized=False, special=True),
    151653: AddedToken("<|vision_end|>", rstrip=False, lstrip=False,
single_word=False, normalized=False, special=True),
```

```
    151654: AddedToken("<|vision_pad|>", rstrip=False, lstrip=False,
single_word=False, normalized=False, special=True),
    151655: AddedToken("<|image_pad|>", rstrip=False, lstrip=False,
single_word=False, normalized=False, special=True),
    151656: AddedToken("<|video_pad|>", rstrip=False, lstrip=False,
single_word=False, normalized=False, special=True),
    151657: AddedToken("<tool_call>", rstrip=False, lstrip=False,
single_word=False, normalized=False, special=False),
    151658: AddedToken("</tool_call>", rstrip=False, lstrip=False,
single_word=False, normalized=False, special=False),
    151659: AddedToken("<|fim_prefix|>", rstrip=False, lstrip=False,
single_word=False, normalized=False, special=False),
    151660: AddedToken("<|fim_middle|>", rstrip=False, lstrip=False,
single_word=False, normalized=False, special=False),
    151661: AddedToken("<|fim_suffix|>", rstrip=False, lstrip=False,
single_word=False, normalized=False, special=False),
    151662: AddedToken("<|fim_pad|>", rstrip=False, lstrip=False,
single_word=False, normalized=False, special=False),
    151663: AddedToken("<|repo_name|>", rstrip=False, lstrip=False,
single_word=False, normalized=False, special=False),
    151664: AddedToken("<|file_sep|>", rstrip=False, lstrip=False,
single_word=False, normalized=False, special=False),
    151665: AddedToken("<tool_response>", rstrip=False, lstrip=False,
single_word=False, normalized=False, special=False),
    151666: AddedToken("</tool_response>", rstrip=False, lstrip=False,
single_word=False, normalized=False, special=False),
    151667: AddedToken("<think>", rstrip=False, lstrip=False,
single_word=False, normalized=False, special=False),
    151668: AddedToken("</think>", rstrip=False, lstrip=False,
single_word=False, normalized=False, special=False),
}
)
```

```python
def process_func(example):
    MAX_LENGTH = 256
    input_ids, attention_mask, labels = [], [], []
    instruction = tokenizer("\n".join(["Human: " + example["instruction"],
example["input"]]).strip() + "\n\nAssistant: ")
    response = tokenizer(example["output"] + tokenizer.eos_token)
    input_ids = instruction["input_ids"] + response["input_ids"]
    attention_mask = instruction["attention_mask"] +
response["attention_mask"]
    labels = [-100] * len(instruction["input_ids"]) + response["input_ids"]
    if len(input_ids) > MAX_LENGTH:
        input_ids = input_ids[:MAX_LENGTH]
        attention_mask = attention_mask[:MAX_LENGTH]
        labels = labels[:MAX_LENGTH]
    return {
        "input_ids": input_ids,
        "attention_mask": attention_mask,
```

```
        "labels": labels
    }
```

```
tokenized_ds = ds.map(process_func, remove_columns=ds.column_names)
tokenized_ds
```

```
Map:   0%|          | 0/26858 [00:00<?, ? examples/s]
Dataset({
    features: ['input_ids', 'attention_mask', 'labels'],
    num_rows: 26858
})
```

```
tokenizer.decode(tokenized_ds[1]["input_ids"])
```

```
'Human: 解释为什么以下分数等同于1/4\n输入：4/16\n\nAssistant: 4/16等于1/4是因为我们可以约分分子分母都除以他们的最大公约数4，得到（4÷4）/ (16÷4）=1/4。分数的约分是用分子和分母除以相同的非零整数，来表示分数的一个相同的值，这因为分数实际上表示了分子除以分母，所以即使两个数同时除以同一个非零整数，分数的值也不会改变。所以4/16 和1/4是两种不同的书写形式，但它们的值相等。<|im_end|>'
```

```
tokenizer.decode(list(filter(lambda x: x != -100, tokenized_ds[1]
["labels"])))
```

```
'4/16等于1/4是因为我们可以约分分子分母都除以他们的最大公约数4，得到（4÷4）/ (16÷4）=1/4。分数的约分是用分子和分母除以相同的非零整数，来表示分数的一个相同的值，这因为分数实际上表示了分子除以分母，所以即使两个数同时除以同一个非零整数，分数的值也不会改变。所以4/16 和1/4是两种不同的书写形式，但它们的值相等。<|im_end|>'
```

Step4 创建模型

```
model = AutoModelForCausalLM.from_pretrained("Qwen/Qwen3-0.6B")
```

```
/usr/local/lib/python3.11/dist-
packages/pydantic/_internal/_generate_schema.py:2225:
UnsupportedFieldAttributeWarning: The 'repr' attribute with value False was
```

```
provided to the `Field()` function, which has no effect in the context it
was used. 'repr' is field-specific metadata, and can only be attached to a
model field using `Annotated` metadata or by assignment. This may have
happened because an `Annotated` type alias using the `type` statement was
used, or if the `Field()` function was attached to a single member of a
union type.
  warnings.warn(
/usr/local/lib/python3.11/dist-
packages/pydantic/_internal/_generate_schema.py:2225:
UnsupportedFieldAttributeWarning: The 'frozen' attribute with value True
was provided to the `Field()` function, which has no effect in the context
it was used. 'frozen' is field-specific metadata, and can only be attached
to a model field using `Annotated` metadata or by assignment. This may have
happened because an `Annotated` type alias using the `type` statement was
used, or if the `Field()` function was attached to a single member of a
union type.
  warnings.warn(
```

```python
for name, parameter in model.named_parameters():
    print(name)
```

```
model.embed_tokens.weight
model.layers.0.self_attn.q_proj.weight
model.layers.0.self_attn.k_proj.weight
model.layers.0.self_attn.v_proj.weight
model.layers.0.self_attn.o_proj.weight
model.layers.0.self_attn.q_norm.weight
model.layers.0.self_attn.k_norm.weight
model.layers.0.mlp.gate_proj.weight
model.layers.0.mlp.up_proj.weight
model.layers.0.mlp.down_proj.weight
model.layers.0.input_layernorm.weight
model.layers.0.post_attention_layernorm.weight
model.layers.1.self_attn.q_proj.weight
model.layers.1.self_attn.k_proj.weight
model.layers.1.self_attn.v_proj.weight
model.layers.1.self_attn.o_proj.weight
model.layers.1.self_attn.q_norm.weight
model.layers.1.self_attn.k_norm.weight
model.layers.1.mlp.gate_proj.weight
model.layers.1.mlp.up_proj.weight
model.layers.1.mlp.down_proj.weight
model.layers.1.input_layernorm.weight
model.layers.1.post_attention_layernorm.weight
model.layers.2.self_attn.q_proj.weight
model.layers.2.self_attn.k_proj.weight
model.layers.2.self_attn.v_proj.weight
model.layers.2.self_attn.o_proj.weight
model.layers.2.self_attn.q_norm.weight
```

```
model.layers.2.self_attn.k_norm.weight
model.layers.2.mlp.gate_proj.weight
model.layers.2.mlp.up_proj.weight
model.layers.2.mlp.down_proj.weight
model.layers.2.input_layernorm.weight
model.layers.2.post_attention_layernorm.weight
model.layers.3.self_attn.q_proj.weight
model.layers.3.self_attn.k_proj.weight
model.layers.3.self_attn.v_proj.weight
model.layers.3.self_attn.o_proj.weight
model.layers.3.self_attn.q_norm.weight
model.layers.3.self_attn.k_norm.weight
model.layers.3.mlp.gate_proj.weight
model.layers.3.mlp.up_proj.weight
model.layers.3.mlp.down_proj.weight
model.layers.3.input_layernorm.weight
model.layers.3.post_attention_layernorm.weight
model.layers.4.self_attn.q_proj.weight
model.layers.4.self_attn.k_proj.weight
model.layers.4.self_attn.v_proj.weight
model.layers.4.self_attn.o_proj.weight
model.layers.4.self_attn.q_norm.weight
model.layers.4.self_attn.k_norm.weight
model.layers.4.mlp.gate_proj.weight
model.layers.4.mlp.up_proj.weight
model.layers.4.mlp.down_proj.weight
model.layers.4.input_layernorm.weight
model.layers.4.post_attention_layernorm.weight
model.layers.5.self_attn.q_proj.weight
model.layers.5.self_attn.k_proj.weight
model.layers.5.self_attn.v_proj.weight
model.layers.5.self_attn.o_proj.weight
model.layers.5.self_attn.q_norm.weight
model.layers.5.self_attn.k_norm.weight
model.layers.5.mlp.gate_proj.weight
model.layers.5.mlp.up_proj.weight
model.layers.5.mlp.down_proj.weight
model.layers.5.input_layernorm.weight
model.layers.5.post_attention_layernorm.weight
model.layers.6.self_attn.q_proj.weight
model.layers.6.self_attn.k_proj.weight
model.layers.6.self_attn.v_proj.weight
model.layers.6.self_attn.o_proj.weight
model.layers.6.self_attn.q_norm.weight
model.layers.6.self_attn.k_norm.weight
model.layers.6.mlp.gate_proj.weight
model.layers.6.mlp.up_proj.weight
model.layers.6.mlp.down_proj.weight
model.layers.6.input_layernorm.weight
model.layers.6.post_attention_layernorm.weight
model.layers.7.self_attn.q_proj.weight
model.layers.7.self_attn.k_proj.weight
model.layers.7.self_attn.v_proj.weight
model.layers.7.self_attn.o_proj.weight
```

```
model.layers.7.self_attn.q_norm.weight
model.layers.7.self_attn.k_norm.weight
model.layers.7.mlp.gate_proj.weight
model.layers.7.mlp.up_proj.weight
model.layers.7.mlp.down_proj.weight
model.layers.7.input_layernorm.weight
model.layers.7.post_attention_layernorm.weight
model.layers.8.self_attn.q_proj.weight
model.layers.8.self_attn.k_proj.weight
model.layers.8.self_attn.v_proj.weight
model.layers.8.self_attn.o_proj.weight
model.layers.8.self_attn.q_norm.weight
model.layers.8.self_attn.k_norm.weight
model.layers.8.mlp.gate_proj.weight
model.layers.8.mlp.up_proj.weight
model.layers.8.mlp.down_proj.weight
model.layers.8.input_layernorm.weight
model.layers.8.post_attention_layernorm.weight
model.layers.9.self_attn.q_proj.weight
model.layers.9.self_attn.k_proj.weight
model.layers.9.self_attn.v_proj.weight
model.layers.9.self_attn.o_proj.weight
model.layers.9.self_attn.q_norm.weight
model.layers.9.self_attn.k_norm.weight
model.layers.9.mlp.gate_proj.weight
model.layers.9.mlp.up_proj.weight
model.layers.9.mlp.down_proj.weight
model.layers.9.input_layernorm.weight
model.layers.9.post_attention_layernorm.weight
model.layers.10.self_attn.q_proj.weight
model.layers.10.self_attn.k_proj.weight
model.layers.10.self_attn.v_proj.weight
model.layers.10.self_attn.o_proj.weight
model.layers.10.self_attn.q_norm.weight
model.layers.10.self_attn.k_norm.weight
model.layers.10.mlp.gate_proj.weight
model.layers.10.mlp.up_proj.weight
model.layers.10.mlp.down_proj.weight
model.layers.10.input_layernorm.weight
model.layers.10.post_attention_layernorm.weight
model.layers.11.self_attn.q_proj.weight
model.layers.11.self_attn.k_proj.weight
model.layers.11.self_attn.v_proj.weight
model.layers.11.self_attn.o_proj.weight
model.layers.11.self_attn.q_norm.weight
model.layers.11.self_attn.k_norm.weight
model.layers.11.mlp.gate_proj.weight
model.layers.11.mlp.up_proj.weight
model.layers.11.mlp.down_proj.weight
model.layers.11.input_layernorm.weight
model.layers.11.post_attention_layernorm.weight
model.layers.12.self_attn.q_proj.weight
model.layers.12.self_attn.k_proj.weight
model.layers.12.self_attn.v_proj.weight
```

```
model.layers.12.self_attn.o_proj.weight
model.layers.12.self_attn.q_norm.weight
model.layers.12.self_attn.k_norm.weight
model.layers.12.mlp.gate_proj.weight
model.layers.12.mlp.up_proj.weight
model.layers.12.mlp.down_proj.weight
model.layers.12.input_layernorm.weight
model.layers.12.post_attention_layernorm.weight
model.layers.13.self_attn.q_proj.weight
model.layers.13.self_attn.k_proj.weight
model.layers.13.self_attn.v_proj.weight
model.layers.13.self_attn.o_proj.weight
model.layers.13.self_attn.q_norm.weight
model.layers.13.self_attn.k_norm.weight
model.layers.13.mlp.gate_proj.weight
model.layers.13.mlp.up_proj.weight
model.layers.13.mlp.down_proj.weight
model.layers.13.input_layernorm.weight
model.layers.13.post_attention_layernorm.weight
model.layers.14.self_attn.q_proj.weight
model.layers.14.self_attn.k_proj.weight
model.layers.14.self_attn.v_proj.weight
model.layers.14.self_attn.o_proj.weight
model.layers.14.self_attn.q_norm.weight
model.layers.14.self_attn.k_norm.weight
model.layers.14.mlp.gate_proj.weight
model.layers.14.mlp.up_proj.weight
model.layers.14.mlp.down_proj.weight
model.layers.14.input_layernorm.weight
model.layers.14.post_attention_layernorm.weight
model.layers.15.self_attn.q_proj.weight
model.layers.15.self_attn.k_proj.weight
model.layers.15.self_attn.v_proj.weight
model.layers.15.self_attn.o_proj.weight
model.layers.15.self_attn.q_norm.weight
model.layers.15.self_attn.k_norm.weight
model.layers.15.mlp.gate_proj.weight
model.layers.15.mlp.up_proj.weight
model.layers.15.mlp.down_proj.weight
model.layers.15.input_layernorm.weight
model.layers.15.post_attention_layernorm.weight
model.layers.16.self_attn.q_proj.weight
model.layers.16.self_attn.k_proj.weight
model.layers.16.self_attn.v_proj.weight
model.layers.16.self_attn.o_proj.weight
model.layers.16.self_attn.q_norm.weight
model.layers.16.self_attn.k_norm.weight
model.layers.16.mlp.gate_proj.weight
model.layers.16.mlp.up_proj.weight
model.layers.16.mlp.down_proj.weight
model.layers.16.input_layernorm.weight
model.layers.16.post_attention_layernorm.weight
model.layers.17.self_attn.q_proj.weight
model.layers.17.self_attn.k_proj.weight
```

```
model.layers.17.self_attn.v_proj.weight
model.layers.17.self_attn.o_proj.weight
model.layers.17.self_attn.q_norm.weight
model.layers.17.self_attn.k_norm.weight
model.layers.17.mlp.gate_proj.weight
model.layers.17.mlp.up_proj.weight
model.layers.17.mlp.down_proj.weight
model.layers.17.input_layernorm.weight
model.layers.17.post_attention_layernorm.weight
model.layers.18.self_attn.q_proj.weight
model.layers.18.self_attn.k_proj.weight
model.layers.18.self_attn.v_proj.weight
model.layers.18.self_attn.o_proj.weight
model.layers.18.self_attn.q_norm.weight
model.layers.18.self_attn.k_norm.weight
model.layers.18.mlp.gate_proj.weight
model.layers.18.mlp.up_proj.weight
model.layers.18.mlp.down_proj.weight
model.layers.18.input_layernorm.weight
model.layers.18.post_attention_layernorm.weight
model.layers.19.self_attn.q_proj.weight
model.layers.19.self_attn.k_proj.weight
model.layers.19.self_attn.v_proj.weight
model.layers.19.self_attn.o_proj.weight
model.layers.19.self_attn.q_norm.weight
model.layers.19.self_attn.k_norm.weight
model.layers.19.mlp.gate_proj.weight
model.layers.19.mlp.up_proj.weight
model.layers.19.mlp.down_proj.weight
model.layers.19.input_layernorm.weight
model.layers.19.post_attention_layernorm.weight
model.layers.20.self_attn.q_proj.weight
model.layers.20.self_attn.k_proj.weight
model.layers.20.self_attn.v_proj.weight
model.layers.20.self_attn.o_proj.weight
model.layers.20.self_attn.q_norm.weight
model.layers.20.self_attn.k_norm.weight
model.layers.20.mlp.gate_proj.weight
model.layers.20.mlp.up_proj.weight
model.layers.20.mlp.down_proj.weight
model.layers.20.input_layernorm.weight
model.layers.20.post_attention_layernorm.weight
model.layers.21.self_attn.q_proj.weight
model.layers.21.self_attn.k_proj.weight
model.layers.21.self_attn.v_proj.weight
model.layers.21.self_attn.o_proj.weight
model.layers.21.self_attn.q_norm.weight
model.layers.21.self_attn.k_norm.weight
model.layers.21.mlp.gate_proj.weight
model.layers.21.mlp.up_proj.weight
model.layers.21.mlp.down_proj.weight
model.layers.21.input_layernorm.weight
model.layers.21.post_attention_layernorm.weight
model.layers.22.self_attn.q_proj.weight
```

```
model.layers.22.self_attn.k_proj.weight
model.layers.22.self_attn.v_proj.weight
model.layers.22.self_attn.o_proj.weight
model.layers.22.self_attn.q_norm.weight
model.layers.22.self_attn.k_norm.weight
model.layers.22.mlp.gate_proj.weight
model.layers.22.mlp.up_proj.weight
model.layers.22.mlp.down_proj.weight
model.layers.22.input_layernorm.weight
model.layers.22.post_attention_layernorm.weight
model.layers.23.self_attn.q_proj.weight
model.layers.23.self_attn.k_proj.weight
model.layers.23.self_attn.v_proj.weight
model.layers.23.self_attn.o_proj.weight
model.layers.23.self_attn.q_norm.weight
model.layers.23.self_attn.k_norm.weight
model.layers.23.mlp.gate_proj.weight
model.layers.23.mlp.up_proj.weight
model.layers.23.mlp.down_proj.weight
model.layers.23.input_layernorm.weight
model.layers.23.post_attention_layernorm.weight
model.layers.24.self_attn.q_proj.weight
model.layers.24.self_attn.k_proj.weight
model.layers.24.self_attn.v_proj.weight
model.layers.24.self_attn.o_proj.weight
model.layers.24.self_attn.q_norm.weight
model.layers.24.self_attn.k_norm.weight
model.layers.24.mlp.gate_proj.weight
model.layers.24.mlp.up_proj.weight
model.layers.24.mlp.down_proj.weight
model.layers.24.input_layernorm.weight
model.layers.24.post_attention_layernorm.weight
model.layers.25.self_attn.q_proj.weight
model.layers.25.self_attn.k_proj.weight
model.layers.25.self_attn.v_proj.weight
model.layers.25.self_attn.o_proj.weight
model.layers.25.self_attn.q_norm.weight
model.layers.25.self_attn.k_norm.weight
model.layers.25.mlp.gate_proj.weight
model.layers.25.mlp.up_proj.weight
model.layers.25.mlp.down_proj.weight
model.layers.25.input_layernorm.weight
model.layers.25.post_attention_layernorm.weight
model.layers.26.self_attn.q_proj.weight
model.layers.26.self_attn.k_proj.weight
model.layers.26.self_attn.v_proj.weight
model.layers.26.self_attn.o_proj.weight
model.layers.26.self_attn.q_norm.weight
model.layers.26.self_attn.k_norm.weight
model.layers.26.mlp.gate_proj.weight
model.layers.26.mlp.up_proj.weight
model.layers.26.mlp.down_proj.weight
model.layers.26.input_layernorm.weight
model.layers.26.post_attention_layernorm.weight
```

```
model.layers.27.self_attn.q_proj.weight
model.layers.27.self_attn.k_proj.weight
model.layers.27.self_attn.v_proj.weight
model.layers.27.self_attn.o_proj.weight
model.layers.27.self_attn.q_norm.weight
model.layers.27.self_attn.k_norm.weight
model.layers.27.mlp.gate_proj.weight
model.layers.27.mlp.up_proj.weight
model.layers.27.mlp.down_proj.weight
model.layers.27.input_layernorm.weight
model.layers.27.post_attention_layernorm.weight
model.norm.weight
```

Lora

PEFT Step1 配置文件

```python
from peft import LoraConfig, TaskType, get_peft_model

config = LoraConfig(task_type=TaskType.CAUSAL_LM, modules_to_save=
["word_embeddings"])
config
```

```
LoraConfig(task_type=<TaskType.CAUSAL_LM: 'CAUSAL_LM'>, peft_type=
<PeftType.LORA: 'LORA'>, auto_mapping=None, base_model_name_or_path=None,
revision=None, inference_mode=False, r=8, target_modules=None,
exclude_modules=None, lora_alpha=8, lora_dropout=0.0, fan_in_fan_out=False,
bias='none', use_rslora=False, modules_to_save=['word_embeddings'],
init_lora_weights=True, layers_to_transform=None, layers_pattern=None,
rank_pattern={}, alpha_pattern={}, megatron_config=None,
megatron_core='megatron.core', trainable_token_indices=None, loftq_config=
{}, eva_config=None, corda_config=None, use_dora=False, use_qalora=False,
qalora_group_size=16, layer_replication=None,
runtime_config=LoraRuntimeConfig(ephemeral_gpu_offload=False),
lora_bias=False)
```

PEFT Step2 创建模型

```python
model = get_peft_model(model, config)
```

```python
config
```

```
LoraConfig(task_type=<TaskType.CAUSAL_LM: 'CAUSAL_LM'>, peft_type=
<PeftType.LORA: 'LORA'>, auto_mapping=None,
base_model_name_or_path='Qwen/Qwen3-0.6B', revision=None,
inference_mode=False, r=8, target_modules={'q_proj', 'v_proj'},
exclude_modules=None, lora_alpha=8, lora_dropout=0.0, fan_in_fan_out=False,
bias='none', use_rslora=False, modules_to_save=['word_embeddings'],
init_lora_weights=True, layers_to_transform=None, layers_pattern=None,
rank_pattern={}, alpha_pattern={}, megatron_config=None,
megatron_core='megatron.core', trainable_token_indices=None, loftq_config=
{}, eva_config=None, corda_config=None, use_dora=False, use_qalora=False,
qalora_group_size=16, layer_replication=None,
runtime_config=LoraRuntimeConfig(ephemeral_gpu_offload=False),
lora_bias=False)
```

```
for name, parameter in model.named_parameters():
    print(name)
```

```
base_model.model.model.embed_tokens.weight
base_model.model.model.layers.0.self_attn.q_proj.base_layer.weight
base_model.model.model.layers.0.self_attn.q_proj.lora_A.default.weight
base_model.model.model.layers.0.self_attn.q_proj.lora_B.default.weight
base_model.model.model.layers.0.self_attn.k_proj.weight
base_model.model.model.layers.0.self_attn.v_proj.base_layer.weight
base_model.model.model.layers.0.self_attn.v_proj.lora_A.default.weight
base_model.model.model.layers.0.self_attn.v_proj.lora_B.default.weight
base_model.model.model.layers.0.self_attn.o_proj.weight
base_model.model.model.layers.0.self_attn.q_norm.weight
base_model.model.model.layers.0.self_attn.k_norm.weight
base_model.model.model.layers.0.mlp.gate_proj.weight
base_model.model.model.layers.0.mlp.up_proj.weight
base_model.model.model.layers.0.mlp.down_proj.weight
base_model.model.model.layers.0.input_layernorm.weight
base_model.model.model.layers.0.post_attention_layernorm.weight
base_model.model.model.layers.1.self_attn.q_proj.base_layer.weight
base_model.model.model.layers.1.self_attn.q_proj.lora_A.default.weight
base_model.model.model.layers.1.self_attn.q_proj.lora_B.default.weight
base_model.model.model.layers.1.self_attn.k_proj.weight
base_model.model.model.layers.1.self_attn.v_proj.base_layer.weight
base_model.model.model.layers.1.self_attn.v_proj.lora_A.default.weight
base_model.model.model.layers.1.self_attn.v_proj.lora_B.default.weight
base_model.model.model.layers.1.self_attn.o_proj.weight
base_model.model.model.layers.1.self_attn.q_norm.weight
base_model.model.model.layers.1.self_attn.k_norm.weight
base_model.model.model.layers.1.mlp.gate_proj.weight
base_model.model.model.layers.1.mlp.up_proj.weight
base_model.model.model.layers.1.mlp.down_proj.weight
base_model.model.model.layers.1.input_layernorm.weight
base_model.model.model.layers.1.post_attention_layernorm.weight
```

```
base_model.model.model.layers.2.self_attn.q_proj.base_layer.weight
base_model.model.model.layers.2.self_attn.q_proj.lora_A.default.weight
base_model.model.model.layers.2.self_attn.q_proj.lora_B.default.weight
base_model.model.model.layers.2.self_attn.k_proj.weight
base_model.model.model.layers.2.self_attn.v_proj.base_layer.weight
base_model.model.model.layers.2.self_attn.v_proj.lora_A.default.weight
base_model.model.model.layers.2.self_attn.v_proj.lora_B.default.weight
base_model.model.model.layers.2.self_attn.o_proj.weight
base_model.model.model.layers.2.self_attn.q_norm.weight
base_model.model.model.layers.2.self_attn.k_norm.weight
base_model.model.model.layers.2.mlp.gate_proj.weight
base_model.model.model.layers.2.mlp.up_proj.weight
base_model.model.model.layers.2.mlp.down_proj.weight
base_model.model.model.layers.2.input_layernorm.weight
base_model.model.model.layers.2.post_attention_layernorm.weight
base_model.model.model.layers.3.self_attn.q_proj.base_layer.weight
base_model.model.model.layers.3.self_attn.q_proj.lora_A.default.weight
base_model.model.model.layers.3.self_attn.q_proj.lora_B.default.weight
base_model.model.model.layers.3.self_attn.k_proj.weight
base_model.model.model.layers.3.self_attn.v_proj.base_layer.weight
base_model.model.model.layers.3.self_attn.v_proj.lora_A.default.weight
base_model.model.model.layers.3.self_attn.v_proj.lora_B.default.weight
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base_model.model.model.layers.3.self_attn.k_norm.weight
base_model.model.model.layers.3.mlp.gate_proj.weight
base_model.model.model.layers.3.mlp.up_proj.weight
base_model.model.model.layers.3.mlp.down_proj.weight
base_model.model.model.layers.3.input_layernorm.weight
base_model.model.model.layers.3.post_attention_layernorm.weight
base_model.model.model.layers.4.self_attn.q_proj.base_layer.weight
base_model.model.model.layers.4.self_attn.q_proj.lora_A.default.weight
base_model.model.model.layers.4.self_attn.q_proj.lora_B.default.weight
base_model.model.model.layers.4.self_attn.k_proj.weight
base_model.model.model.layers.4.self_attn.v_proj.base_layer.weight
base_model.model.model.layers.4.self_attn.v_proj.lora_A.default.weight
base_model.model.model.layers.4.self_attn.v_proj.lora_B.default.weight
base_model.model.model.layers.4.self_attn.o_proj.weight
base_model.model.model.layers.4.self_attn.q_norm.weight
base_model.model.model.layers.4.self_attn.k_norm.weight
base_model.model.model.layers.4.mlp.gate_proj.weight
base_model.model.model.layers.4.mlp.up_proj.weight
base_model.model.model.layers.4.mlp.down_proj.weight
base_model.model.model.layers.4.input_layernorm.weight
base_model.model.model.layers.4.post_attention_layernorm.weight
base_model.model.model.layers.5.self_attn.q_proj.base_layer.weight
base_model.model.model.layers.5.self_attn.q_proj.lora_A.default.weight
base_model.model.model.layers.5.self_attn.q_proj.lora_B.default.weight
base_model.model.model.layers.5.self_attn.k_proj.weight
base_model.model.model.layers.5.self_attn.v_proj.base_layer.weight
base_model.model.model.layers.5.self_attn.v_proj.lora_A.default.weight
base_model.model.model.layers.5.self_attn.v_proj.lora_B.default.weight
base_model.model.model.layers.5.self_attn.o_proj.weight
base_model.model.model.layers.5.self_attn.q_norm.weight
```

```
base_model.model.model.layers.5.self_attn.k_norm.weight
base_model.model.model.layers.5.mlp.gate_proj.weight
base_model.model.model.layers.5.mlp.up_proj.weight
base_model.model.model.layers.5.mlp.down_proj.weight
base_model.model.model.layers.5.input_layernorm.weight
base_model.model.model.layers.5.post_attention_layernorm.weight
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base_model.model.model.layers.6.self_attn.q_proj.lora_B.default.weight
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base_model.model.model.layers.6.self_attn.v_proj.base_layer.weight
base_model.model.model.layers.6.self_attn.v_proj.lora_A.default.weight
base_model.model.model.layers.6.self_attn.v_proj.lora_B.default.weight
base_model.model.model.layers.6.self_attn.o_proj.weight
base_model.model.model.layers.6.self_attn.q_norm.weight
base_model.model.model.layers.6.self_attn.k_norm.weight
base_model.model.model.layers.6.mlp.gate_proj.weight
base_model.model.model.layers.6.mlp.up_proj.weight
base_model.model.model.layers.6.mlp.down_proj.weight
base_model.model.model.layers.6.input_layernorm.weight
base_model.model.model.layers.6.post_attention_layernorm.weight
base_model.model.model.layers.7.self_attn.q_proj.base_layer.weight
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base_model.model.model.layers.7.self_attn.k_proj.weight
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base_model.model.model.layers.8.self_attn.v_proj.lora_A.default.weight
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base_model.model.model.layers.8.mlp.gate_proj.weight
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base_model.model.model.layers.8.input_layernorm.weight
base_model.model.model.layers.8.post_attention_layernorm.weight
base_model.model.model.layers.9.self_attn.q_proj.base_layer.weight
base_model.model.model.layers.9.self_attn.q_proj.lora_A.default.weight
base_model.model.model.layers.9.self_attn.q_proj.lora_B.default.weight
```

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base_model.model.model.layers.9.self_attn.k_proj.weight
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base_model.model.model.layers.11.input_layernorm.weight
base_model.model.model.layers.11.post_attention_layernorm.weight
base_model.model.model.layers.12.self_attn.q_proj.base_layer.weight
base_model.model.model.layers.12.self_attn.q_proj.lora_A.default.weight
base_model.model.model.layers.12.self_attn.q_proj.lora_B.default.weight
base_model.model.model.layers.12.self_attn.k_proj.weight
base_model.model.model.layers.12.self_attn.v_proj.base_layer.weight
base_model.model.model.layers.12.self_attn.v_proj.lora_A.default.weight
base_model.model.model.layers.12.self_attn.v_proj.lora_B.default.weight
base_model.model.model.layers.12.self_attn.o_proj.weight
base_model.model.model.layers.12.self_attn.q_norm.weight
base_model.model.model.layers.12.self_attn.k_norm.weight
base_model.model.model.layers.12.mlp.gate_proj.weight
base_model.model.model.layers.12.mlp.up_proj.weight
```

```
base_model.model.model.layers.12.mlp.down_proj.weight
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base_model.model.model.layers.12.post_attention_layernorm.weight
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base_model.model.model.layers.14.self_attn.v_proj.lora_B.default.weight
base_model.model.model.layers.14.self_attn.o_proj.weight
base_model.model.model.layers.14.self_attn.q_norm.weight
base_model.model.model.layers.14.self_attn.k_norm.weight
base_model.model.model.layers.14.mlp.gate_proj.weight
base_model.model.model.layers.14.mlp.up_proj.weight
base_model.model.model.layers.14.mlp.down_proj.weight
base_model.model.model.layers.14.input_layernorm.weight
base_model.model.model.layers.14.post_attention_layernorm.weight
base_model.model.model.layers.15.self_attn.q_proj.base_layer.weight
base_model.model.model.layers.15.self_attn.q_proj.lora_A.default.weight
base_model.model.model.layers.15.self_attn.q_proj.lora_B.default.weight
base_model.model.model.layers.15.self_attn.k_proj.weight
base_model.model.model.layers.15.self_attn.v_proj.base_layer.weight
base_model.model.model.layers.15.self_attn.v_proj.lora_A.default.weight
base_model.model.model.layers.15.self_attn.v_proj.lora_B.default.weight
base_model.model.model.layers.15.self_attn.o_proj.weight
base_model.model.model.layers.15.self_attn.q_norm.weight
base_model.model.model.layers.15.self_attn.k_norm.weight
base_model.model.model.layers.15.mlp.gate_proj.weight
base_model.model.model.layers.15.mlp.up_proj.weight
base_model.model.model.layers.15.mlp.down_proj.weight
base_model.model.model.layers.15.input_layernorm.weight
base_model.model.model.layers.15.post_attention_layernorm.weight
base_model.model.model.layers.16.self_attn.q_proj.base_layer.weight
base_model.model.model.layers.16.self_attn.q_proj.lora_A.default.weight
base_model.model.model.layers.16.self_attn.q_proj.lora_B.default.weight
base_model.model.model.layers.16.self_attn.k_proj.weight
base_model.model.model.layers.16.self_attn.v_proj.base_layer.weight
base_model.model.model.layers.16.self_attn.v_proj.lora_A.default.weight
```

```
    base_model.model.model.layers.16.self_attn.v_proj.lora_B.default.weight
    base_model.model.model.layers.16.self_attn.o_proj.weight
    base_model.model.model.layers.16.self_attn.q_norm.weight
    base_model.model.model.layers.16.self_attn.k_norm.weight
    base_model.model.model.layers.16.mlp.gate_proj.weight
    base_model.model.model.layers.16.mlp.up_proj.weight
    base_model.model.model.layers.16.mlp.down_proj.weight
    base_model.model.model.layers.16.input_layernorm.weight
    base_model.model.model.layers.16.post_attention_layernorm.weight
    base_model.model.model.layers.17.self_attn.q_proj.base_layer.weight
    base_model.model.model.layers.17.self_attn.q_proj.lora_A.default.weight
    base_model.model.model.layers.17.self_attn.q_proj.lora_B.default.weight
    base_model.model.model.layers.17.self_attn.k_proj.weight
    base_model.model.model.layers.17.self_attn.v_proj.base_layer.weight
    base_model.model.model.layers.17.self_attn.v_proj.lora_A.default.weight
    base_model.model.model.layers.17.self_attn.v_proj.lora_B.default.weight
    base_model.model.model.layers.17.self_attn.o_proj.weight
    base_model.model.model.layers.17.self_attn.q_norm.weight
    base_model.model.model.layers.17.self_attn.k_norm.weight
    base_model.model.model.layers.17.mlp.gate_proj.weight
    base_model.model.model.layers.17.mlp.up_proj.weight
    base_model.model.model.layers.17.mlp.down_proj.weight
    base_model.model.model.layers.17.input_layernorm.weight
    base_model.model.model.layers.17.post_attention_layernorm.weight
    base_model.model.model.layers.18.self_attn.q_proj.base_layer.weight
    base_model.model.model.layers.18.self_attn.q_proj.lora_A.default.weight
    base_model.model.model.layers.18.self_attn.q_proj.lora_B.default.weight
    base_model.model.model.layers.18.self_attn.k_proj.weight
    base_model.model.model.layers.18.self_attn.v_proj.base_layer.weight
    base_model.model.model.layers.18.self_attn.v_proj.lora_A.default.weight
    base_model.model.model.layers.18.self_attn.v_proj.lora_B.default.weight
    base_model.model.model.layers.18.self_attn.o_proj.weight
    base_model.model.model.layers.18.self_attn.q_norm.weight
    base_model.model.model.layers.18.self_attn.k_norm.weight
    base_model.model.model.layers.18.mlp.gate_proj.weight
    base_model.model.model.layers.18.mlp.up_proj.weight
    base_model.model.model.layers.18.mlp.down_proj.weight
    base_model.model.model.layers.18.input_layernorm.weight
    base_model.model.model.layers.18.post_attention_layernorm.weight
    base_model.model.model.layers.19.self_attn.q_proj.base_layer.weight
    base_model.model.model.layers.19.self_attn.q_proj.lora_A.default.weight
    base_model.model.model.layers.19.self_attn.q_proj.lora_B.default.weight
    base_model.model.model.layers.19.self_attn.k_proj.weight
    base_model.model.model.layers.19.self_attn.v_proj.base_layer.weight
    base_model.model.model.layers.19.self_attn.v_proj.lora_A.default.weight
    base_model.model.model.layers.19.self_attn.v_proj.lora_B.default.weight
    base_model.model.model.layers.19.self_attn.o_proj.weight
    base_model.model.model.layers.19.self_attn.q_norm.weight
    base_model.model.model.layers.19.self_attn.k_norm.weight
    base_model.model.model.layers.19.mlp.gate_proj.weight
    base_model.model.model.layers.19.mlp.up_proj.weight
    base_model.model.model.layers.19.mlp.down_proj.weight
    base_model.model.model.layers.19.input_layernorm.weight
    base_model.model.model.layers.19.post_attention_layernorm.weight
```

```
base_model.model.model.layers.20.self_attn.q_proj.base_layer.weight
base_model.model.model.layers.20.self_attn.q_proj.lora_A.default.weight
base_model.model.model.layers.20.self_attn.q_proj.lora_B.default.weight
base_model.model.model.layers.20.self_attn.k_proj.weight
base_model.model.model.layers.20.self_attn.v_proj.base_layer.weight
base_model.model.model.layers.20.self_attn.v_proj.lora_A.default.weight
base_model.model.model.layers.20.self_attn.v_proj.lora_B.default.weight
base_model.model.model.layers.20.self_attn.o_proj.weight
base_model.model.model.layers.20.self_attn.q_norm.weight
base_model.model.model.layers.20.self_attn.k_norm.weight
base_model.model.model.layers.20.mlp.gate_proj.weight
base_model.model.model.layers.20.mlp.up_proj.weight
base_model.model.model.layers.20.mlp.down_proj.weight
base_model.model.model.layers.20.input_layernorm.weight
base_model.model.model.layers.20.post_attention_layernorm.weight
base_model.model.model.layers.21.self_attn.q_proj.base_layer.weight
base_model.model.model.layers.21.self_attn.q_proj.lora_A.default.weight
base_model.model.model.layers.21.self_attn.q_proj.lora_B.default.weight
base_model.model.model.layers.21.self_attn.k_proj.weight
base_model.model.model.layers.21.self_attn.v_proj.base_layer.weight
base_model.model.model.layers.21.self_attn.v_proj.lora_A.default.weight
base_model.model.model.layers.21.self_attn.v_proj.lora_B.default.weight
base_model.model.model.layers.21.self_attn.o_proj.weight
base_model.model.model.layers.21.self_attn.q_norm.weight
base_model.model.model.layers.21.self_attn.k_norm.weight
base_model.model.model.layers.21.mlp.gate_proj.weight
base_model.model.model.layers.21.mlp.up_proj.weight
base_model.model.model.layers.21.mlp.down_proj.weight
base_model.model.model.layers.21.input_layernorm.weight
base_model.model.model.layers.21.post_attention_layernorm.weight
base_model.model.model.layers.22.self_attn.q_proj.base_layer.weight
base_model.model.model.layers.22.self_attn.q_proj.lora_A.default.weight
base_model.model.model.layers.22.self_attn.q_proj.lora_B.default.weight
base_model.model.model.layers.22.self_attn.k_proj.weight
base_model.model.model.layers.22.self_attn.v_proj.base_layer.weight
base_model.model.model.layers.22.self_attn.v_proj.lora_A.default.weight
base_model.model.model.layers.22.self_attn.v_proj.lora_B.default.weight
base_model.model.model.layers.22.self_attn.o_proj.weight
base_model.model.model.layers.22.self_attn.q_norm.weight
base_model.model.model.layers.22.self_attn.k_norm.weight
base_model.model.model.layers.22.mlp.gate_proj.weight
base_model.model.model.layers.22.mlp.up_proj.weight
base_model.model.model.layers.22.mlp.down_proj.weight
base_model.model.model.layers.22.input_layernorm.weight
base_model.model.model.layers.22.post_attention_layernorm.weight
base_model.model.model.layers.23.self_attn.q_proj.base_layer.weight
base_model.model.model.layers.23.self_attn.q_proj.lora_A.default.weight
base_model.model.model.layers.23.self_attn.q_proj.lora_B.default.weight
base_model.model.model.layers.23.self_attn.k_proj.weight
base_model.model.model.layers.23.self_attn.v_proj.base_layer.weight
base_model.model.model.layers.23.self_attn.v_proj.lora_A.default.weight
base_model.model.model.layers.23.self_attn.v_proj.lora_B.default.weight
base_model.model.model.layers.23.self_attn.o_proj.weight
base_model.model.model.layers.23.self_attn.q_norm.weight
```

```
base_model.model.model.layers.23.self_attn.k_norm.weight
base_model.model.model.layers.23.mlp.gate_proj.weight
base_model.model.model.layers.23.mlp.up_proj.weight
base_model.model.model.layers.23.mlp.down_proj.weight
base_model.model.model.layers.23.input_layernorm.weight
base_model.model.model.layers.23.post_attention_layernorm.weight
base_model.model.model.layers.24.self_attn.q_proj.base_layer.weight
base_model.model.model.layers.24.self_attn.q_proj.lora_A.default.weight
base_model.model.model.layers.24.self_attn.q_proj.lora_B.default.weight
base_model.model.model.layers.24.self_attn.k_proj.weight
base_model.model.model.layers.24.self_attn.v_proj.base_layer.weight
base_model.model.model.layers.24.self_attn.v_proj.lora_A.default.weight
base_model.model.model.layers.24.self_attn.v_proj.lora_B.default.weight
base_model.model.model.layers.24.self_attn.o_proj.weight
base_model.model.model.layers.24.self_attn.q_norm.weight
base_model.model.model.layers.24.self_attn.k_norm.weight
base_model.model.model.layers.24.mlp.gate_proj.weight
base_model.model.model.layers.24.mlp.up_proj.weight
base_model.model.model.layers.24.mlp.down_proj.weight
base_model.model.model.layers.24.input_layernorm.weight
base_model.model.model.layers.24.post_attention_layernorm.weight
base_model.model.model.layers.25.self_attn.q_proj.base_layer.weight
base_model.model.model.layers.25.self_attn.q_proj.lora_A.default.weight
base_model.model.model.layers.25.self_attn.q_proj.lora_B.default.weight
base_model.model.model.layers.25.self_attn.k_proj.weight
base_model.model.model.layers.25.self_attn.v_proj.base_layer.weight
base_model.model.model.layers.25.self_attn.v_proj.lora_A.default.weight
base_model.model.model.layers.25.self_attn.v_proj.lora_B.default.weight
base_model.model.model.layers.25.self_attn.o_proj.weight
base_model.model.model.layers.25.self_attn.q_norm.weight
base_model.model.model.layers.25.self_attn.k_norm.weight
base_model.model.model.layers.25.mlp.gate_proj.weight
base_model.model.model.layers.25.mlp.up_proj.weight
base_model.model.model.layers.25.mlp.down_proj.weight
base_model.model.model.layers.25.input_layernorm.weight
base_model.model.model.layers.25.post_attention_layernorm.weight
base_model.model.model.layers.26.self_attn.q_proj.base_layer.weight
base_model.model.model.layers.26.self_attn.q_proj.lora_A.default.weight
base_model.model.model.layers.26.self_attn.q_proj.lora_B.default.weight
base_model.model.model.layers.26.self_attn.k_proj.weight
base_model.model.model.layers.26.self_attn.v_proj.base_layer.weight
base_model.model.model.layers.26.self_attn.v_proj.lora_A.default.weight
base_model.model.model.layers.26.self_attn.v_proj.lora_B.default.weight
base_model.model.model.layers.26.self_attn.o_proj.weight
base_model.model.model.layers.26.self_attn.q_norm.weight
base_model.model.model.layers.26.self_attn.k_norm.weight
base_model.model.model.layers.26.mlp.gate_proj.weight
base_model.model.model.layers.26.mlp.up_proj.weight
base_model.model.model.layers.26.mlp.down_proj.weight
base_model.model.model.layers.26.input_layernorm.weight
base_model.model.model.layers.26.post_attention_layernorm.weight
base_model.model.model.layers.27.self_attn.q_proj.base_layer.weight
base_model.model.model.layers.27.self_attn.q_proj.lora_A.default.weight
base_model.model.model.layers.27.self_attn.q_proj.lora_B.default.weight
```

```
base_model.model.model.layers.27.self_attn.k_proj.weight
base_model.model.model.layers.27.self_attn.v_proj.base_layer.weight
base_model.model.model.layers.27.self_attn.v_proj.lora_A.default.weight
base_model.model.model.layers.27.self_attn.v_proj.lora_B.default.weight
base_model.model.model.layers.27.self_attn.o_proj.weight
base_model.model.model.layers.27.self_attn.q_norm.weight
base_model.model.model.layers.27.self_attn.k_norm.weight
base_model.model.model.layers.27.mlp.gate_proj.weight
base_model.model.model.layers.27.mlp.up_proj.weight
base_model.model.model.layers.27.mlp.down_proj.weight
base_model.model.model.layers.27.input_layernorm.weight
base_model.model.model.layers.27.post_attention_layernorm.weight
base_model.model.model.norm.weight
```

```
model.print_trainable_parameters()
```

```
trainable params: 1,146,880 || all params: 597,196,800 || trainable%:
0.1920
```

Step5 配置训练参数

```
args = TrainingArguments(
    output_dir="./chatbot",
    per_device_train_batch_size=2,
    gradient_accumulation_steps=8,
    logging_steps=10,
    num_train_epochs=1,
    report_to="none"
)
```

Step6 创建训练器

```
trainer = Trainer(
    model=model,
    args=args,
    tokenizer=tokenizer,
    train_dataset=tokenized_ds,
    data_collator=DataCollatorForSeq2Seq(tokenizer=tokenizer,
padding=True),
)
```

```
/tmp/ipykernel_38/1988846643.py:1: FutureWarning: `tokenizer` is deprecated
and will be removed in version 5.0.0 for `Trainer.__init__`. Use
`processing_class` instead.
  trainer = Trainer(
No label_names provided for model class `PeftModelForCausalLM`. Since
`PeftModel` hides base models input arguments, if label_names is not given,
label_names can't be set automatically within `Trainer`. Note that empty
label_names list will be used instead.
```

Step7 模型训练

```
trainer.train()
```

10 2.561900 20 2.563500 30 2.493000 40 2.269800 50 2.271200 60 2.185600 70 2.213600 80 2.167400 90 2.100500 100 2.020000 110 2.037200 120 2.030400 130 2.033100 140 1.988400 150 2.058200 160 1.983200 170 1.855600 180 1.982200 190 2.010100 200 2.044200 210 1.949300 220 2.029500 230 1.915900 240 1.945900 250 2.084000 260 1.945300 270 1.996100 280 1.985800 290 2.035900 300 1.923400 310 1.963400 320 1.905600 330 1.924900 340 1.937800 350 1.905900 360 1.987600 370 2.052800 380 1.904700 390 1.905900 400 2.078600 410 1.921500 420 2.000800 430 1.919400 440 2.023400 450 1.987500 460 1.928700 470 2.019000 480 1.926400 490 1.945600 500 1.872600 510 1.962000 520 1.885100 530 1.881900 540 1.937000 550 1.874400 560 1.801500 570 2.035800 580 1.851300 590 2.027800 600 1.946000 610 1.949300 620 2.003200 630 1.840900 640 1.954100 650 1.916900 660 1.908600 670 1.913300 680 1.879400 690 1.889400 700 1.923300 710 1.900500 720 2.025600 730 1.897100 740 1.834300 750 1.994300 760 1.900700 770 1.995500 780 1.946800 790 1.908500 800 1.999700 810 1.850800 820 1.841900 830 1.899800 840 1.915600 850 1.853900 860 1.855600 870 1.889900 880 1.896900 890 1.922200 900 1.894100 910 1.948100 920 2.016200 930 1.938500 940 1.963800 950 2.039800 960 1.872200 970 1.841800 980 1.927700 990 2.009800 1000 1.965000 1010 1.903200 1020 1.824100 1030 1.954500 1040 1.823800 1050 1.884900 1060 2.039300 1070 1.946300 1080 1.862700 1090 1.836600 1100 1.901400 1110 1.928900 1120 1.871400 1130 1.845200 1140 1.826000 1150 1.907400 1160 1.860300 1170 2.002500 1180 1.946100 1190 1.794700 1200 1.882500 1210 2.011200 1220 1.953300 1230 1.801600 1240 1.997200 1250 1.860300 1260 1.956000 1270 1.972500 1280 1.904300 1290 1.880400 1300 2.020600 1310 1.941700 1320 1.907400 1330 1.911400 1340 1.986100 1350 1.857300 1360 1.940300 1370 2.023900 1380 1.842300 1390 1.878000 1400 1.898700 1410 1.857100 1420 1.966600 1430 1.903800 1440 2.007800 1450 1.836400 1460 1.922900 1470 1.889100 1480 1.891400 1490 1.944400 1500 1.959100 1510 1.898900 1520 1.854600 1530 1.905900 1540 1.941700 1550 1.902300 1560 1.860600 1570 1.840300 1580 1.945200 1590 1.909100 1600 1.878100 1610 1.885100 1620 1.852300 1630 1.894900 1640 1.950200 1650 1.877100 1660 1.891100 1670 1.892300

```
TrainOutput(global_step=1679, training_loss=1.949883185517866, metrics=
{'train_runtime': 2115.3691, 'train_samples_per_second': 12.697,
'train_steps_per_second': 0.794, 'total_flos': 7152758508355584.0,
'train_loss': 1.949883185517866, 'epoch': 1.0})
```

Step8 模型推理

```
model = model.cuda()
ipt = tokenizer("Human: {}\n{}".format("考试有哪些技巧？", "").strip() + 
"\n\nAssistant: ", return_tensors="pt").to(model.device)
tokenizer.decode(model.generate(**ipt, max_length=128, do_sample=True)[0],
skip_special_tokens=True)
```

```
'Human: 考试有哪些技巧？\n\nAssistant: 考试技巧有很多，以下是一些常见的技巧，可以帮助你更好地准备和应对考试：\n\n1. 制定计划：根据你所学的知识和考试内容，制定一个详细的学习计划，包括每天的学习时间、复习计划和目标。\n\n2. 做好复习：复习是考试成功的必要条件，你可以利用笔记、练习题和错题本来复习你的知识。同时，可以使用一些复习方法，如背诵、口述、记忆和归纳等。\n\n3. 多做题：通过大量练习，你可以提高解题能力'
```

```
model = model.cuda()
ipt = tokenizer("Human: {}\n{}".format("你叫什么名字？", "").strip() + 
"\n\nAssistant: ", return_tensors="pt").to(model.device)
tokenizer.decode(model.generate(**ipt, max_length=128, do_sample=True)[0],
skip_special_tokens=True)
```

```
'Human: 你叫什么名字？\n\nAssistant: 我叫我叫"我"，是一个虚拟助手，可以为您提供各种帮助和支持。'
```

```
model = model.cuda()
ipt = tokenizer("Human: {}\n{}".format("你知道江西科技学院吗？", "").strip() + 
"\n\nAssistant: ", return_tensors="pt").to(model.device)
tokenizer.decode(model.generate(**ipt, max_length=128, do_sample=True)[0],
skip_special_tokens=True)
```

```
'Human: 你知道江西科技学院吗？\n\nAssistant: 江西科技学院位于江西省南昌市，是一所综合性本科院校。它是由江西省人民政府发起设立的，成立于2005年，拥有江西省科技厅主管的职能。学院的学科门类广泛，涵盖计算机、数学、物理、化学、生物、医学、教育学等多门，设有多个国家级和省级重点实验室，拥有丰富的科研成果和良好的教学资源。江西科技学院注重实践教学，重视学生创新能力的培养，致力于为社会培养高素质、高技能的科技人才。'
```

Step9 拓展尝试

9.1 增加 epoch

```
model = AutoModelForCausalLM.from_pretrained("Qwen/Qwen3-0.6B")
model = get_peft_model(model, config)
```

```
args = TrainingArguments(
    output_dir="./chatbot",
    per_device_train_batch_size=2,
    gradient_accumulation_steps=8,
    logging_steps=10,
    num_train_epochs=3,
    report_to="none"
)
```

```
trainer = Trainer(
    model=model,
    args=args,
    tokenizer=tokenizer,
    train_dataset=tokenized_ds,
    data_collator=DataCollatorForSeq2Seq(tokenizer=tokenizer,
padding=True),
)
```

```
/tmp/ipykernel_38/1988846643.py:1: FutureWarning: `tokenizer` is deprecated
and will be removed in version 5.0.0 for `Trainer.__init__`. Use
`processing_class` instead.
  trainer = Trainer(
No label_names provided for model class `PeftModelForCausalLM`. Since
`PeftModel` hides base models input arguments, if label_names is not given,
label_names can't be set automatically within `Trainer`. Note that empty
label_names list will be used instead.
```

```
trainer.train()
```

Step Training Loss 10 2.560800 20 2.560000 30 2.486500 40 2.262800 50 2.262700 60 2.178900 70 2.203800 80 2.156000 90 2.087500 100 2.012300 110 2.035700 120 2.029000 130 2.032300 140 1.987200 150 2.056600 160 1.981600 170 1.853600 180 1.980800 190 2.008100 200 2.043000 210 1.948500 220 2.028100 230 1.914100 240 1.944700 250 2.081800 260 1.942800 270 1.993700 280 1.984400 290 2.033700 300 1.920800 310 1.961600 320 1.903700 330 1.922000 340 1.936200 350 1.903500 360 1.985100 370 2.050300 380 1.902300 390 1.903300 400 2.075600 410 1.919500 420 1.998200 430 1.915400 440 2.020700 450 1.985100 460 1.926200 470 2.016200 480 1.921300 490 1.943600 500 1.868700 510 1.958500 520 1.882900 530 1.878800 540 1.933200 550 1.871500 560 1.798800 570 2.033600 580 1.847400 590 2.023700 600 1.941200 610 1.946900 620 2.000400 630 1.837300 640

1.951800 650 1.913600 660 1.905800 670 1.909800 680 1.876200 690 1.885000 700 1.919500 710
1.896300 720 2.021600 730 1.893200 740 1.830800 750 1.989800 760 1.896300 770 1.991100 780
1.944000 790 1.904800 800 1.995100 810 1.847500 820 1.836800 830 1.895500 840 1.911800 850
1.850500 860 1.849800 870 1.884200 880 1.890400 890 1.917200 900 1.887700 910 1.942200 920
2.011900 930 1.931600 940 1.959200 950 2.033500 960 1.867200 970 1.835300 980 1.922600 990
2.002300 1000 1.959400 1010 1.896600 1020 1.819600 1030 1.948500 1040 1.817300 1050 1.881500 1060
2.031600 1070 1.939600 1080 1.854900 1090 1.831700 1100 1.895700 1110 1.921600 1120 1.864200 1130
1.837800 1140 1.819400 1150 1.899600 1160 1.852700 1170 1.994200 1180 1.938700 1190 1.788100 1200
1.873400 1210 2.003800 1220 1.946400 1230 1.795000 1240 1.989200 1250 1.851900 1260 1.949200 1270
1.966000 1280 1.895600 1290 1.871000 1300 2.012500 1310 1.934000 1320 1.898300 1330 1.903100 1340
1.977300 1350 1.846500 1360 1.930900 1370 2.013200 1380 1.832000 1390 1.869100 1400 1.889100 1410
1.849400 1420 1.958000 1430 1.894300 1440 1.996400 1450 1.827500 1460 1.913500 1470 1.878000 1480
1.880500 1490 1.934500 1500 1.950700 1510 1.890100 1520 1.844300 1530 1.893400 1540 1.929800 1550
1.890900 1560 1.850600 1570 1.829800 1580 1.934700 1590 1.898000 1600 1.870000 1610 1.874600 1620
1.840200 1630 1.883000 1640 1.937700 1650 1.863600 1660 1.878200 1670 1.881700 1680 1.912500 1690
1.938900 1700 1.924100 1710 1.927600 1720 1.853600 1730 1.842900 1740 1.906600 1750 1.930700 1760
1.843900 1770 1.815200 1780 1.921300 1790 1.991000 1800 1.911800 1810 1.914300 1820 1.896000 1830
1.871300 1840 2.010900 1850 1.860100 1860 1.949800 1870 1.840700 1880 1.934900 1890 1.988600 1900
1.877900 1910 1.893300 1920 1.862300 1930 1.972900 1940 1.853100 1950 1.896500 1960 1.857600 1970
1.926400 1980 1.946300 1990 1.930200 2000 1.789300 2010 1.951400 2020 1.864600 2030 1.850200 2040
1.897600 2050 1.851800 2060 1.885800 2070 1.883300 2080 1.863000 2090 1.961100 2100 1.917200 2110
1.960700 2120 1.813800 2130 1.917800 2140 1.818500 2150 1.872700 2160 1.835000 2170 1.799700 2180
1.921700 2190 1.836100 2200 2.007500 2210 1.925600 2220 1.981600 2230 1.836100 2240 1.862500 2250
1.875400 2260 1.886800 2270 1.859000 2280 1.860900 2290 1.869500 2300 1.888500 2310 1.868900 2320
1.797600 2330 1.875300 2340 1.797400 2350 1.942200 2360 1.898000 2370 1.893900 2380 1.936200 2390
1.823900 2400 1.860200 2410 1.887700 2420 1.815000 2430 1.927800 2440 1.853700 2450 1.867000 2460
1.943700 2470 1.850400 2480 1.893000 2490 1.822400 2500 1.918100 2510 2.008200 2520 1.819500 2530
1.823500 2540 2.006900 2550 1.974200 2560 1.855000 2570 2.015200 2580 1.939300 2590 1.873400 2600
1.850700 2610 1.922400 2620 1.876000 2630 1.843200 2640 1.833700 2650 1.895300 2660 1.787300 2670
1.868400 2680 1.931000 2690 1.910500 2700 1.983000 2710 1.838700 2720 1.871900 2730 1.959600 2740
1.907900 2750 1.804200 2760 1.796900 2770 1.920800 2780 1.933700 2790 1.811000 2800 1.770900 2810
1.927500 2820 1.941200 2830 1.971400 2840 1.833400 2850 1.943600 2860 1.914600 2870 1.966200 2880
1.888300 2890 1.734000 2900 1.903100 2910 1.938600 2920 1.984100 2930 1.880400 2940 2.035400 2950
1.746600 2960 1.876200 2970 1.827700 2980 1.867500 2990 1.837200 3000 1.984400 3010 1.933000 3020
1.947700 3030 1.861500 3040 1.724600 3050 1.880500 3060 1.827800 3070 1.885800 3080 1.945400 3090
1.901200 3100 1.873900 3110 1.943500 3120 1.862600 3130 1.914300 3140 1.869000 3150 1.935100 3160
1.920600 3170 1.762600 3180 1.908100 3190 1.764700 3200 1.920800 3210 1.813600 3220 1.800400 3230
1.790000 3240 1.842700 3250 1.786100 3260 1.823100 3270 1.811300 3280 1.837100 3290 1.827300 3300
1.780500 3310 1.883300 3320 1.940000 3330 1.812900 3340 1.894100 3350 1.797600 3360 1.817900 3370
1.908100 3380 1.912900 3390 1.731500 3400 1.784900 3410 1.913900 3420 1.883700 3430 1.905700 3440
1.911400 3450 1.779900 3460 1.810000 3470 1.846200 3480 1.857400 3490 1.795700 3500 1.689400 3510
1.867700 3520 1.936300 3530 1.945800 3540 1.783000 3550 1.840900 3560 1.779800 3570 1.875600 3580
1.899100 3590 1.843300 3600 1.920000 3610 1.844600 3620 1.942700 3630 1.973100 3640 1.871300 3650
1.917500 3660 1.771200 3670 1.860600 3680 1.904400 3690 1.839200 3700 1.892000 3710 1.870500 3720
1.838100 3730 1.840500 3740 2.067900 3750 1.777700 3760 1.811300 3770 1.872000 3780 1.905900 3790
1.835700 3800 1.803800 3810 1.866400 3820 1.811100 3830 1.841800 3840 1.845200 3850 1.771400 3860

1.815000 3870 1.808000 3880 1.850800 3890 1.860100 3900 1.880000 3910 1.863200 3920 1.892800 3930 1.874900 3940 1.862200 3950 1.885000 3960 1.963100 3970 1.867600 3980 1.813000 3990 1.761900 4000 1.890100 4010 1.792000 4020 1.900600 4030 1.793200 4040 1.934400 4050 1.923900 4060 1.969400 4070 2.002300 4080 1.952200 4090 1.879300 4100 1.822800 4110 1.902500 4120 1.860200 4130 1.805100 4140 1.888700 4150 1.950100 4160 1.759000 4170 1.882800 4180 1.842400 4190 1.873500 4200 1.795100 4210 1.824300 4220 1.790000 4230 1.899900 4240 1.923000 4250 1.880000 4260 1.940400 4270 1.827900 4280 1.891300 4290 1.936200 4300 1.829900 4310 1.842100 4320 1.780700 4330 1.919900 4340 1.806200 4350 1.903500 4360 1.882400 4370 1.820000 4380 1.957700 4390 1.814500 4400 1.739600 4410 1.846000 4420 1.766400 4430 1.930400 4440 1.815200 4450 1.826800 4460 1.885000 4470 1.828600 4480 1.874100 4490 1.929000 4500 1.881600 4510 1.859500 4520 1.886900 4530 1.866100 4540 1.899000